

Question	Answer	Marks
9(a)	diffraction is characteristic of wave behaviour so shows that electrons can behave like waves	B1
9(b)	$qV = \frac{1}{2}mv^2$	C1
	$p = mv$	C1
	$p = m \times \sqrt{(2qV / m)}$ $= \sqrt{(2qVm)}$	A1
9(c)	(electrons have) greater momentum <u>so</u> smaller (de Broglie) wavelength	B1
	fringes become closer together	B1
9(d)(i)	straight line with positive gradient	B1
	line with positive gradient passing through the origin	B1
9(d)(ii)	Planck constant	B1

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